Day-4 Oct-03

# Intro To CLI

CLI- Command Line Interface.
We'll use CLI to configure ~~Network~~ CISCO devices.

We can also use GUI but most network Engineers prefer to use the CLI over GUI.

## # How to connect Cisco device to configure it with CLI?

↳ Dherai method xan, among them one is via console port.
Ethernet जोडे जस्तै laptop/PC बाट wire ल्याउने अनि console Port मा जोड्ने । ~~यदि~~

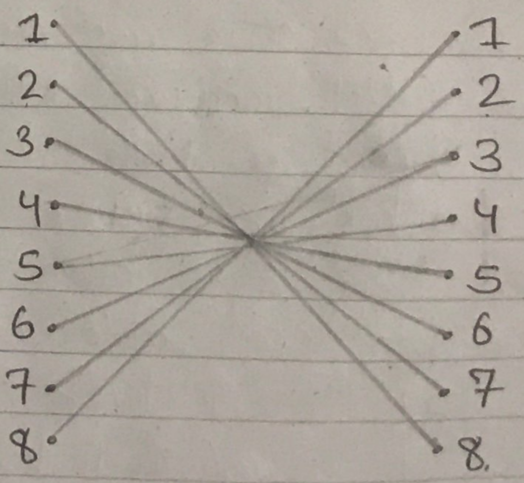
यदि RJ45 port छ भने, एकातिर RJ45, ~~र~~ connector र अर्को तिर DB9 connector वाला चाहिन्छ। आजकल को laptop मा db9 connector हुने ठाउँ नहुँदा ~~so~~ adapter might be needed.

त्यो RJ45-dB9 connector cable को actual name chai rollover cable ho hai.

Ethernet UTP cable हरु मा जस्तै यसमा पनि 8 वटा Pin हुन्छ, which are joined as below.

So, once we've connected our computer to device, we'll need to use a terminal emulator to access CLI.

A popular tool for this is PuTTY. (putty.org).
~~Putty config~~



# Putty Terminal.

Connection type मा serial choose garni. ani open garni, & well we will get connected to CLI. Default settings मा connect गर्न सकिन्छ, तर अझै जान्न मन भए त्यो Category मा serial मा click गर्दा Right side मा आउँछ, एउटा dialog box, जसबाट ~~default~~ settings change गर्न सकिन्छ.

These settings ~~will~~ match the default settings, so probably we'll never ~~to~~ have to change them, but it's good to be aware of them & remember them for exams.

Speed (baud rate) - 9600 bps (bits/sec)
Data bits - 8
Stop bit - 1 [For each 8 bit of data 1 stop bit is sent] to mark end of 8 bit
Parity - none (checks errors)
flow control - none (controls flow of data from transmitter to receiver)

# When terminal opens.

When ~~e~~ we click open in putty terminal, the terminal opens & we get greeted. ~~We~~ In tutorial we'll use Router CLI to configure Router.

User - Exec Mode.

- When we first enter CLI, we'll by default be ~~in~~ in 'user EXEC mode'.
- User exec mode will be denote by '>' greater than sign. Router >
- It is very ~~to~~ limited.
- Users can look at some things, but can't make any changes to the configuration.
- Also called user-mode

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## #. Privileged EXEC MODE.

- User-exec mode मा 'enable' lekhesi we'll reach to privileged exec mode
- It will be denoted by '#' dik eg: Router #
- Provides complete access to view the device's configuration, restart the device, etc.
- In this mode one can't change config, but can change time on device, save the configuration file. etc.

### Some commands of USER EXEC & PRIVileged Exec-Mode.

| User-Exec | Privi-Exec. |
|---|---|
| <1-99> | • User Exec's को & सबै |
| connect | • mkdir |
| disable | • copy |
| enable | • debug. |
| disconnect. | • erase. |
| ping | • rmdir |
| show | • setup |
| telnet | • undebug |
| traceroute | • vlan |
| terminal | • write. |
| logout | • reload. |

NOTE: Terminal मा 'en' lekhera tab click garo bhane autocomplete hunxa.
en मात्र लेखेर enter [illegible] enable नै command run hunxa

To view available command ली run the command: ? (question mark)

Con ? जस्तै linux मा tab thichda available options dinthyo, similarly yesko cli मा ~~pres~~ press '?' questionmark like con? गर्यो भने connect config option dinchha

## Global configuratio Mode.

- To enter this mode go-to priv-exec mode. Then type "configure terminal"
- ~~term~~ Hostname (in our case Router) will be followed by '(config)' which denotes we are in global config mode.

# enable-password.

security risk ले गर्दा Priv-exec ~~...~~ mode मा जान Password चाहिने बनाउनुपर्छ। To do so ~~+~~ global config mode मा जाने

- Then type 'enable password'

NOTE: ~~pass~~ '?' को अगाडि no space then it shows us all possible completions of that word.
'?' को अगाडि space दियो भने त्यो line मा use गर्न मिल्ने ~~...~~ अरु possible command दिन्छ।

- Passwords are case sensitive.
- <cr> indicates no further options available.

# Running-config / Startup-config.

There are two separate configuration files kept on the device at once.

- Running-config: ~~the curr~~ the current,active config file on the device. As we enter the commands, we edit the active configuration
- Startup-config: the configuration file that will be loaded upon the restart of device

Global-config mode मा huda wa aru mode मा हुँदा
If we wanna run privileged exec mode command type do infront
e.g do show running-config

- Use 'show running-config' command on ~~global~~ global config mode to view the running config file.

- Similary 'show startup-config' to see startup config file.

# Saving configuratio

There are 3 ways to save running-config to make it the startup config. All these 3 ~~ca~~ are executed from privileged-exec mode.

- The first is 'write'. It prints 'OK' indicating config was saved
- Second method is 'write memory'. Same as first method.
- Third method is copy running-config startup-config.
  It copies running config file to startup-config file.

# Service Password-encryption.

- In global configuration, use 'service password-encryption' command.
  (यसो गरेसी अरुले config file check garda password ~~hidden~~ ~~basxa~~ encrypt bhyera basxa).

# Enable Secret.

- The more secure method is to use 'enable secret your-password' command

Enable secret र enable password both use गरेको छ भने enable secret will get more priority

- Always use enable-secret instead of enable-password.

# Cancelling or deleting command.

- Use no infront of our config.

Eg.. no service password-encryption. Euta
यस्तो गरेपछि future मा save garya password हरू
encrypted hunnan. तर पहिला encrypt bhyera basya thyo
tyo encrypted nai hunxa, tyo decrypted hunna.

## QUIZ.

Q.1) What kind of cable is used to ~~convert~~ connect to a CISCO device via RJ45 Console port?

[✓ a) Roll-over] b) Cross-over c) USb-cable.

Q.2) You type enable to enter privileged exec mode on your Cisco router, however the password you enter is not accepted. What could be prob

a) Service password-encryption is enable
b) service password-encryption is disable
[✓ c) Caps Lock is on.]

Q.3) What is the most secure method to protect access to priveleged exec mode

[✓ a) enable secret command] b) enable password command
c) enable password cmnd with service password-encryption.

Q.4) If both enable-pass & enable-secret config are enabled, what will happe
when you use enable to enter Privileged-exec mode?
a) you must enter enable password, followed by enable secret
b) you must enter enable password only
[✓ c) you must enter enable secret only.]

Q.5) You used 'conf t' commd to enter global config mode. What is full length ver
a) configuration time [✓ b) configure terminal] c) configuration termina